

PROJECT SETUP AND CONFIGURATION

AI StudyBuddy – An Intelligent Personalized Learning Assistant

1. Software Requirements

- Operating System: Windows 10/11
- Frontend: HTML, CSS, JavaScript / React.js
- Backend: Python + Flask / FastAPI
- Database: MongoDB
- AI API: Google Gemini API / OpenAI API
- Code Editor: Visual Studio Code
- API Testing: Postman
- Version Control: Git & GitHub

2. Create Project Folder

```
AI-StudyBuddy/  
■  
■■■ frontend/  
■■■ backend/  
■■■ models/  
■■■ routes/  
■■■ uploads/  
■■■ static/  
■■■ templates/  
■■■ database/  
■■■ .env  
■■■ requirements.txt  
■■■ README.md
```

3. Backend Setup

Create a Python virtual environment:

```
python -m venv venv
```

Activate it on Windows:

```
venv\Scripts\activate
```

Install the required packages:

```
pip install flask pymongo python-dotenv  
pip install google-generativeai  
pip install PyMuPDF
```

Save the dependencies:

```
pip freeze > requirements.txt
```

4. MongoDB Configuration

Database Name: AI_StudyBuddy

Suggested collections:

```
students
study_materials
topics
ai_chats
questions
quizzes
quiz_attempts
progress
study_plans
flashcards
```

Example **.env** configuration:

```
MONGO_URI=mongodb://localhost:27017/
DATABASE_NAME=AI_StudyBuddy
GEMINI_API_KEY=your_api_key_here
```

5. Run the Project

Start the backend:

```
python app.py
```

Open the application in your browser:

```
http://127.0.0.1:5000
```

6. Configuration Flow

Install Python → Create Project → Create Virtual Environment → Install Packages → Configure MongoDB → Add AI API Key → Run Flask Server → Open Browser